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Hyperbolic Part-Whole Segmentation

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Abstract

Semantic segmentation typically focuses on classification at a single predefined granularity, neglecting the inherent hierarchical structure of physical entities. Yet, objects naturally decompose into parts and subparts, mirroring human visual perception. To bridge this gap, we introduce a hyperbolic prototypical segmentation framework capable of simultaneously representing multiple granularity levels within a unified embedding space. Leveraging hyperbolic geometry’s capacity to model hierarchies efficiently, we propose embedding class prototypes within the Poincaré ball. We introduce a tree-aware prototype initialization strategy and a distortion- p loss that together yield significantly improved hierarchical embeddings. Furthermore, we derive an optimized formulation of the hyperbolic distance function, enabling tractable inference for dense prediction tasks. A shared transformer encoder paired with separate hyperbolic heads allows efficient multi-level segmentation from a single model. In a different configuration, the model can produce coarse-level predictions without any supervision beyond fine-grained labels, exploiting the semantic alignment of all prototypes on the manifold. Experiments on the SubPartImageNet dataset advance state-of-the-art mIoU at *subpart* and *part* levels by 9% and 11% respectively, and demonstrate remarkable zero-shot capabilities, while using a model two orders of magnitude smaller than prior work.¹

¹Our full code is available at <https://github.com/mikhail-vlasenko/hyp-segmentation>

Chapter 1

Introduction

Segmentation is done at the single, predefined level of granularity: object level. However, this approach largely ignores the fact that real-world objects possess inherent hierarchical structures. For instance, animals can be decomposed into body parts, which in turn consist of smaller sub-parts like eyes or paws. This compositional nature is not just physical; research in cognitive psychology underscores that human visual perception inherently organizes information hierarchically (Biederman, 1987), interpreting scenes in terms of wholes and their constituent parts. However, current segmentation methodologies largely ignore this intrinsic hierarchical composition of most entities, limiting their descriptive power and practical applicability. Aligning computational methods with this cognitive structure promises to enhance the interpretability and effectiveness of segmentation models.

This thesis moves beyond simply acknowledging these hierarchies and argues that the design of the hierarchy itself is a critical modeling choice. The most intuitive hierarchy is not always the most effective for a given task, and we show that reorganizing these relationships can lead to superior performance for specific use cases.

Recently, significant advancements have been made in hierarchical representation learning and segmentation, notably through the introduction of the SubPartImageNet (SPIN) dataset by Myers-Dean et al. (2024), which uniquely annotates natural images across three granularity levels: *wholes*, *parts*, and *subparts*. We provide an example image with annotations in Figure 1.1. While the authors have performed evaluations on the dataset, they did not explore cross-level training and prediction and largely ignored the hierarchical relations between the classes.

The majority of existing approaches operate fully in the Euclidean space, which inherently lacks the capacity to efficiently represent hierarchical relationships due to geometric constraints. To overcome this limitation, hyperbolic embedding spaces have emerged as a compelling alternative due to their exponential growth properties and suitability for naturally modeling hierarchical structures (Mettes et al., 2024).

To effectively leverage this geometry, we adopt prototypical learning, an approach that represents classes using exemplar vectors called prototypes. In this paradigm, the model makes predictions by measuring the distance between an input feature and these class prototypes in the embedding space (Jetley et al., 2015; Snell et al., 2017; Mettes et al., 2019). Our

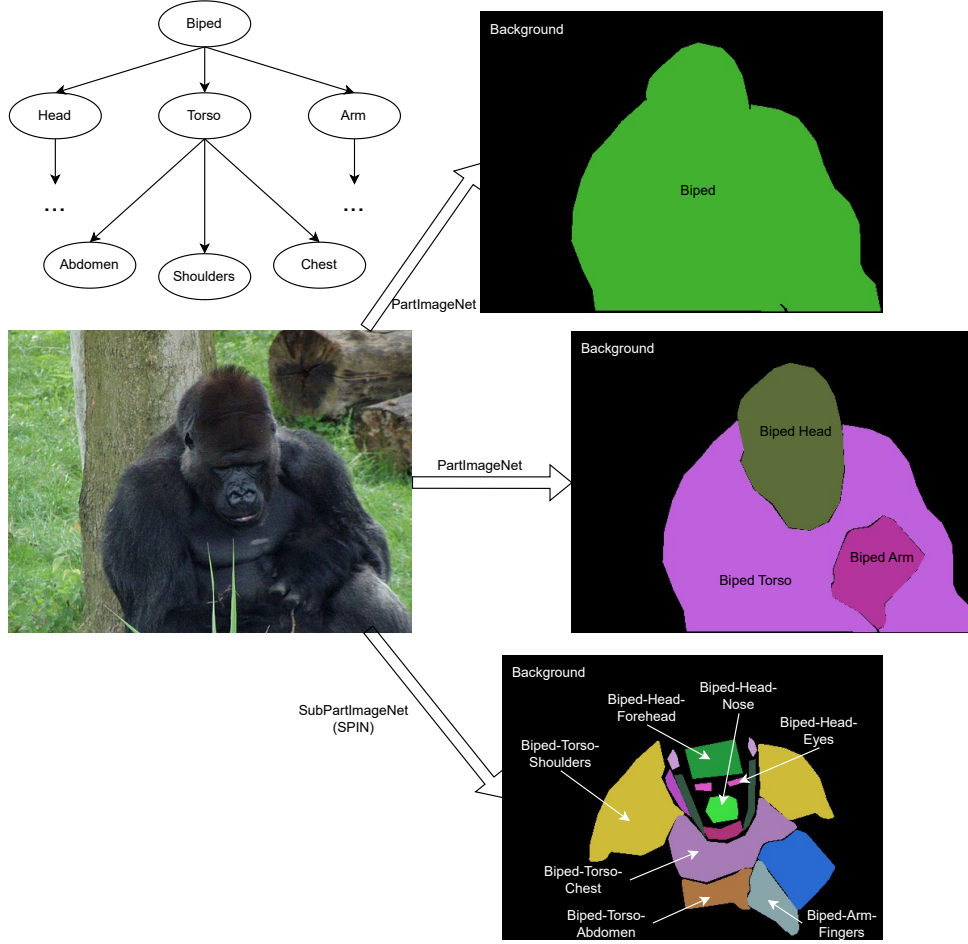


Figure 1.1: Example from the SPIN dataset illustrating its multi-level hierarchical annotations, and a subtree of the suggested class hierarchy.

method adapts this concept for hierarchical semantic segmentation within the hyperbolic space, enabled by a novel reformulation of hyperbolic distance computation. Each semantic class at every hierarchical level is represented by a prototype embedded in the hyperbolic manifold, with its position determined through an iterative, graph-distance-based optimization. This integrated approach provides notable benefits, including computational efficiency, improved training stability, and the flexibility to manipulate prototypes without retraining any part of the model.

This thesis makes the following contributions:

1. We propose a novel multi-head architecture leveraging multiple hyperbolic embeddings simultaneously, each tailored to different hierarchical granularities.
2. We introduce two improvements to the construction of class prototypes within hyperbolic spaces: one for initialization, and one for optimization.
3. We reformulate the hyperbolic distance computation to dramatically increase algorithmic efficiency, enabling us to design the first hyperbolic prototypical learning method for a dense prediction task. This enables novel capabilities such as multi-level predictions from one model output, and vastly improves zero-shot performance.
4. We achieve state-of-the-art mean Intersection-over-Union (mIoU) on the SPIN dataset, demonstrating the superior segmentation performance of hyperbolic methods.

Chapter 2

Related Work

This chapter reviews the literature relevant to our work, placing our contributions in a broader context by first discussing the evolution of general semantic segmentation, then narrowing our focus to the more specific challenge of part-based segmentation, and finally reviewing the application of hyperbolic geometry in machine learning.

2.1 Segmentation

Semantic segmentation, the task of assigning a class label to every pixel in an image, is one of the key problems in computer vision. The field was revolutionized by the introduction of Fully Convolutional Networks (FCNs) by J. Long et al. (2015), who pioneered end-to-end training for dense prediction tasks by replacing fully-connected layers with convolutional ones.

A dominant paradigm that emerged soon after is the encoder-decoder architecture. Models like U-Net (Ronneberger et al., 2015) and SegNet (Badrinarayanan et al., 2017) employ a contracting path (encoder) to capture semantic context and a symmetric expanding path (decoder) to recover fine-grained spatial information. In particular, U-Net introduced skip connections between the encoder and decoder, which proved crucial for preserving high-resolution details. To further enhance contextual understanding without sacrificing spatial resolution, the DeepLab family of models introduced dilated convolutions (L.-C. Chen et al., 2017, 2018). By inserting gaps in convolutional kernels, these models can effectively expand their receptive field, leading to significant performance gains on benchmarks like PASCAL VOC (Everingham et al., 2010) and Cityscapes (Cordts et al., 2016).

Later, the success of Transformers in natural language processing has inspired their application to vision tasks. For this thesis, we use a transformer-based SegFormer model by Xie et al. (2021). The choice is further discussed and motivated in Section 5.4.

More recently, the rise of large-scale foundation models for segmentation led to the Segment Anything Model (SAM) (Kirillov et al., 2023), which trained on an exceptionally large dataset of over one billion masks. The current iteration of this model, SAM 2 (Ravi et al., 2024), goes further in this direction, creating a unified model for segmenting anything in both images and videos. Notably, the data engine of SAM 2 is designed to collect masks for constituent

parts of the objects, acknowledging the need for a more granular understanding. Despite their extraordinary power and flexibility, these models primarily treat segmentation as a general visual grouping task. Their understanding of semantic hierarchies — the knowledge that a leg is part of a biped — remains implicit, learned as a statistical correlation from the vast data rather than being an explicit component of the model’s representational space. Our work directly addresses this gap by building a geometric prior for such relationships, aiming for a model that not only segments, but also structurally understands the composition of objects.

2.2 Part-Based Segmentation

While general segmentation focuses on entire objects, part-based segmentation aims for a more granular decomposition of wholes into components. This increased granularity is crucial for applications requiring detailed object understanding, such as fine-grained recognition and robotic manipulation, and is supported by cognitive science theories on human object perception (Biederman, 1987). Early progress in this area was enabled by the creation of specialized datasets like PASCAL-Part (X. Chen et al., 2014) and the more recent, large-scale Part-ImageNet (J. He et al., 2022). Initial approaches often involved adapting standard segmentation architectures to these datasets, effectively treating each part as an independent class without explicitly modeling its relationship to the parent object (Gonzalez-Garcia et al., 2018).

The recent introduction of the SubPartImageNet (SPIN) dataset marked a significant milestone, providing the first large-scale dataset with dense annotations across three levels of a part-whole hierarchy: *wholes*, *parts*, and *subparts* (Myers-Dean et al., 2024). This has enabled the benchmarking of powerful vision-language models like GLaMM (Rasheed et al., 2024) and LISA (Lai et al., 2024) on a multi-level segmentation task. However, these models suffer from the same fundamental limitation as the foundation models discussed previously. In contrast, our work embeds the class hierarchy directly into the projection space via prototypical learning (Snell et al., 2017; Boney & Ilin, 2017; Mettes et al., 2019).

2.3 Hyperbolic Machine Learning

The challenge of faithfully representing hierarchical data in continuous vector spaces has led researchers to explore non-Euclidean geometries. Hyperbolic manifolds, in particular, have gained prominence due to their natural ability to embed tree-like structures with significantly lower distortion than their Euclidean counterparts (Krioukov et al., 2010; Nickel & Kiela, 2017; Sala et al., 2018; Mettes et al., 2024). The Poincaré ball is one of the most common models of hyperbolic geometry used in machine learning due to its convenient conformal structure (van Spengler, Berkhout, & Mettes, 2023).

The pioneering work of Nickel and Kiela (2017) demonstrated the effectiveness of learning graph node embeddings in the Poincaré ball, showing significant improvements over Euclidean methods for representing complex networks with latent hierarchies. Since then, hyperbolic

methods have been successfully applied to various domains. In natural language processing, they have been used for learning word hierarchies from text corpora (Tifrea et al., 2018), modeling logical entailment (Ganea et al., 2018b), and generating sentence representations (Le et al., 2019). In other areas, they have proven effective for recommender systems (Vinh Tran et al., 2020) and in computer vision for tasks like few-shot learning (Krioukov et al., 2010) and action recognition (T. Long et al., 2020).

Recently, a few works have adapted hyperbolic geometry for semantic segmentation (Atigh et al., 2022; Weber et al., 2024). These methods employ a Hyperbolic Multinomial Logistic Regression classifier (Ganea et al., 2018b). However, these works did not leverage the space to explicitly model a class hierarchy, instead treating classes as independent entities. Further, such an approach produces a model for a fixed set of classes, which severely limits its potential for zero-shot segmentation.

Our work addresses these limitations by departing from complex hyperplane-based classifiers. We introduce the first hyperbolic prototypical learning framework for dense prediction, adapting the idea of hyperbolic prototypical deep learning (Mettes et al., 2019; Kasarla et al., 2022; Pal et al., 2024) to semantic segmentation.

Chapter 3

Preliminaries

This section introduces the core concepts of hyperbolic geometry and its application in machine learning, which are useful for understanding our approach.

3.1 Hyperbolic Spaces

Hyperbolic spaces are a type of non-Euclidean geometry characterized by a constant negative curvature. Unlike Euclidean space, which has zero curvature, the volume of a hyperbolic space grows exponentially with its radius (Cannon et al., 1997). This exponential growth property makes hyperbolic spaces exceptionally well-suited for embedding hierarchical or tree-like data structures with significantly lower distortion than in Euclidean spaces of the same dimension (Nickel & Kiela, 2017; Mettes et al., 2024).

In machine learning, one of the commonly used models for hyperbolic geometry is the Poincaré ball model. We define the d -dimensional Poincaré ball with curvature $c < 0$ as the open unit ball in $\mathbb{R}^d$:

$$\mathbb{B}^d = \{\mathbf{x} \in \mathbb{R}^d : \|\mathbf{x}\| < 1\}.$$

The geometry within this ball is distorted such that the distance between points grows rapidly as they approach the boundary of the ball $\|\mathbf{x}\| = 1$. Consequently, the boundary is considered to be “at infinity”. This structure provides a natural way to represent hierarchies: general, high-level concepts can be placed near the origin (the center of the ball), while more specific, low-level concepts (or leaf nodes) are positioned closer to the boundary.

3.1.1 Exponential Map

The exponential map is used to project vectors from a Euclidean space into the hyperbolic manifold. The exponential map at the origin, $\exp_0^c(\mathbf{v})$, maps a vector $\mathbf{v}$ from the tangent space at the origin (which is a Euclidean space) to a point on the manifold (Ganea et al., 2018b):

$$\exp_0^c(\mathbf{v}) = \tanh(\sqrt{-c}\|\mathbf{v}\|) \frac{\mathbf{v}}{\sqrt{-c}\|\mathbf{v}\|}. \quad (3.1)$$

This operation is crucial for integrating hyperbolic geometry with standard neural network architectures, as it provides a way to map network outputs onto the Poincaré ball.

3.1.2 Möbius Addition

Möbius addition is a fundamental operation that generalizes the standard vector addition to hyperbolic space, maintaining consistency with the hyperbolic geometry. The operation, denoted as $\oplus_c$ in a manifold with curvature c is defined as:

$$x \oplus_c y = \frac{(1 + 2c\langle x, y \rangle + c|y|^2)x + (1 - c|x|^2)y}{1 + 2c\langle x, y \rangle + c^2|x|^2|y|^2}. \quad (3.2)$$

This operation is central to the geometry of the Poincaré ball. Unlike Euclidean addition, it is not commutative ($\mathbf{x} \oplus_c \mathbf{y} \neq \mathbf{y} \oplus_c \mathbf{x}$) nor associative ($\mathbf{x} \oplus_c (\mathbf{y} \oplus_c \mathbf{z}) \neq (\mathbf{x} \oplus_c \mathbf{y}) \oplus_c \mathbf{z}$). Nevertheless, the origin $\mathbf{0}$ acts as the identity element ($\mathbf{x} \oplus_c \mathbf{0} = \mathbf{x}$), and every point $\mathbf{x}$ has a unique inverse $-\mathbf{x}$ such that $\mathbf{x} \oplus_c (-\mathbf{x}) = \mathbf{0}$. Crucially, Möbius addition ensures that the result of combining two vectors always remains within the ball, preserving the manifold’s structure.¹

3.1.3 Hyperbolic Distances

The distance between two points $\mathbf{x}, \mathbf{y} \in \mathbb{B}^d$ is not the standard Euclidean distance but is instead defined by the geodesic path connecting them on the curved manifold. In the Poincaré ball model, this distance can be computed using Möbius addition:

$$d_c(\mathbf{x}, \mathbf{y}) = \frac{2}{\sqrt{-c}} \tanh^{-1} \left(\sqrt{-c} \| -\mathbf{x} \oplus_c \mathbf{y} \| \right).$$

Here, the term $-\mathbf{x} \oplus_c \mathbf{y}$ can be interpreted as the “Möbius difference” or a translation of $\mathbf{y}$ by $-\mathbf{x}$. The distance function scales the Euclidean norm of this resulting vector to account for the curvature of the space. As points move away from the origin and approach the boundary of the ball, their norms approach 1. The $\tanh^{-1}$ function maps these values towards infinity, capturing the exponential expansion of the space and the rapidly increasing distances between points near the boundary. This property is precisely what allows the Poincaré ball to accurately represent the large distances between far leaves in a tree.

3.2 Hyperbolic MLR

Multinomial Logistic Regression (MLR) is a standard method for multi-class classification. Its generalization to hyperbolic space, Hyperbolic MLR, provides a way to define decision boundaries on the hyperbolic manifold (Ganea et al., 2018b).

¹Euclidean addition can produce vectors outside of the hyperbolic manifold. For example, adding the two vectors $\mathbf{x} = (0.99, 0)$ and $\mathbf{y} = (0, 0.99)$ with the standard operation yields $\mathbf{x} + \mathbf{y} = (0.99, 0.99)$, which lies outside the Poincaré ball since $\|\mathbf{x} + \mathbf{y}\| = \sqrt{0.99^2 + 0.99^2} \approx 1.4 > 1$.

A hyperbolic hyperplane is the generalization of a Euclidean hyperplane. It can be defined as the set of points equidistant from two given points or, more formally, as the projection of a linear subspace from a tangent space back onto the manifold. Hyperbolic MLR then defines classification logits based on the signed hyperbolic distance of a point to these hyperplanes. This provides a decision boundary that respects the structure of the hyperbolic space.

However, despite its geometric elegance, the Hyperbolic MLR framework has significant practical drawbacks that limit its scalability and stability. The core issue stems from its reliance on two learnable parameters for each class. This approach is inherently parameter-inefficient, especially for tasks with many categories. More critically, the reliance on on-manifold learnable parameters is the source of significant training instability. During training, gradient-based updates can push these parameters toward the boundary of the Poincaré ball. As a parameter’s norm approaches 1, the denominators in the distance and gradient computations approach zero. This leads to severe floating-point inaccuracies, numerical overflow, and exploding gradients (Chami et al., [2020](#)).

Chapter 4

Method

This chapter outlines the architectural and algorithmic components of our proposed prototypical hyperbolic segmentation framework. Our approach combines hierarchical prototype embeddings, efficient hyperbolic distance computations, and prototypical classification to enable dense, multi-level semantic segmentation that respects part-whole relationships between classes.

We begin by obtaining a class hierarchy in as a graph and, leveraging a two-stage optimization process tailored to hierarchical distances, construct hyperbolic prototypes that represent nodes of that hierarchy. These prototypes are then integrated into a segmentation pipeline, enabling prototype-based classification on a Poincaré ball. To support segmentation-scale inference, we introduce a critical optimization for computing hyperbolic distances efficiently across millions of pixels. Finally, we detail our approach for multi-level predictions, zero-shot evaluation, and the shared encoder strategy that facilitates scalable deployment across different granularity levels in the hierarchy.

4.1 Improved Hyperbolic Hierarchical Prototypes

To encode the part-whole hierarchy, represented as a directed acyclic graph $\mathcal{G} = (V, E)$, we construct a learnable embedding for each node $v \in V$ in the d -dimensional Poincaré ball $\mathbb{B}^d$. This hyperbolic geometry is well-suited for capturing hierarchical relationships between classes with low distortion.

Our procedure consists of three stages:

1. Initialization of the learnable embeddings.
2. Pretraining that adapts the original Poincaré embeddings loss (Nickel & Kiela, 2017).
3. Fine-tuning with a distortion loss tailored to the graph distances in $\mathcal{G}$.

4.1.1 Tree-Aware Initialization

Previous methods (Nickel & Kiela, 2017) initialize embeddings from a uniform distribution $\mathcal{U}(-0.001, 0.001)$. However, we observe that this approach yields poor structural representa-

tion, particularly in lower-dimensional spaces. We deduce that early in training, embeddings organize into suboptimal topologies. Subsequent incremental updates are unable to swap subtrees effectively, as the intermediate states of the swap form a ridge in the loss landscape.

To illustrate this phenomenon, we construct a to-scale minimal example in which a local embedding swap would reduce the overall distortion loss. [Figure 4.1](#) presents the distortion loss landscape for moving the embedding marked by the purple star. While placing the starred embedding at $(-0.03, 0.84)$ on the Poincaré disk would achieve a smaller loss, a transition there would require passing through a ridge around $(0.03, 0.84)$, which is an implausible trajectory for the optimizing SGD. While increasing the embedding dimension partially mitigates this issue as the ridge becomes less extreme, it does not fundamentally address the problem of poor initial topology.

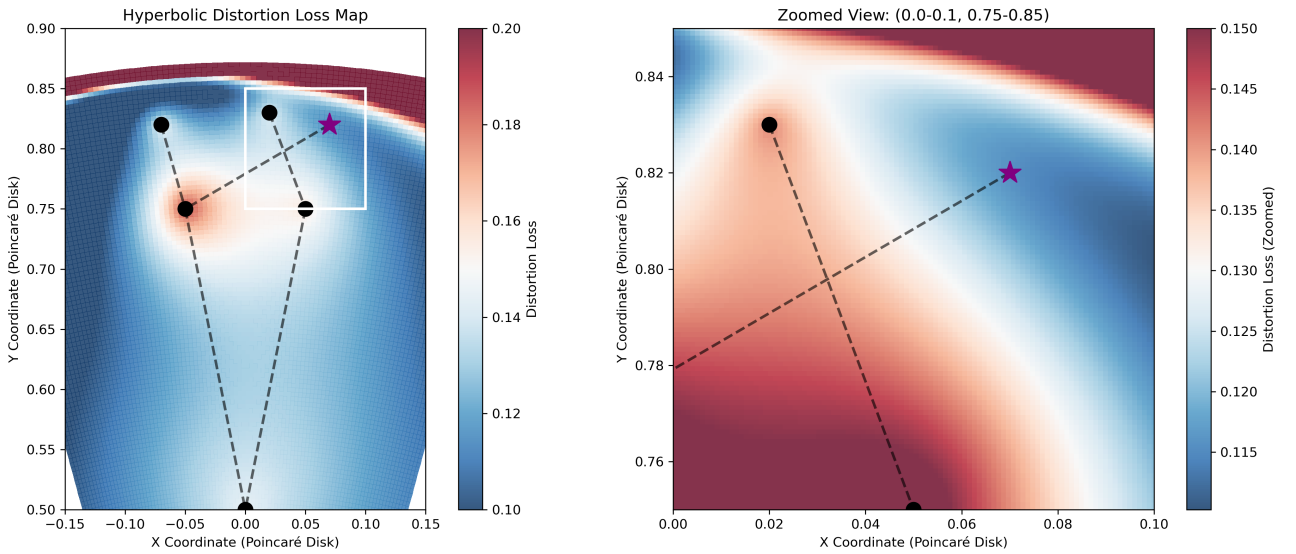


Figure 4.1: Hyperbolic distortion loss landscape for the embedding marked by the purple star. **Left:** landscape of a sector on the Poincaré disk. **Right:** zoomed view of the region outlined in white. A narrow ridge of high loss (orange-white) separates the current position from the lower-loss basin, illustrating why gradient descent struggles to swap subtrees.

As such, we propose an improved initialization strategy that leverages the known hierarchical structure directly. Specifically, embeddings are initialized by traversing the hierarchy depth-first from the root, and placing nodes in a circular arrangement within the Poincaré ball by their traversal order. In higher-dimensional spaces, the first two dimensions adhere to this circular structure, while additional dimensions are initialized from the previously discussed uniform distribution.

We then employ the stochastic approach of (Nickel & Kiela, 2017) and iteratively optimize the embeddings with Riemannian SGD (Béginneul & Ganeva, 2018) on a batch of edges. Mathematically, the loss can be expressed as:

$$\mathcal{L}_{\text{PE}} = -\frac{1}{|B|} \sum_{(u,v) \in B} \log \frac{\exp(-d_{\mathbb{B}}(\mathbf{z}_u, \mathbf{z}_v))}{\sum_{v'} \exp(-d_{\mathbb{B}}(\mathbf{z}_u, \mathbf{z}_{v'}))},$$

where u, v are indices of source and target nodes for an edge; $\mathbf{z}_u, \mathbf{z}_v$ are Poincaré ball embeddings

of nodes u and v ; $d_{\mathbb{B}}$ denotes the hyperbolic distance; and B is a batch of positive edges augmented with sampled negative examples.

As noted by Ganea et al. (2018a), this approach produces embeddings very close to the boundary of the Poincaré ball, so we rescale them by 0.8 to stabilize the optimization in the next phase.

4.1.2 Distortion- p Loss

After rescaling, we fine-tune the embeddings using a distortion loss that aligns hyperbolic distances with target distances d_{uv}^{target} derived from shortest paths in the hierarchy graph $\mathcal{G}$:

$$\mathcal{L}_{\text{dist}} = \frac{1}{|B|} \sum_{(u,v) \in B} \left| \frac{d_{\mathbb{B}}(\mathbf{z}_u, \mathbf{z}_v) - d_{uv}^{\text{target}}}{d_{uv}^{\text{target}}} \right|^p,$$

where p is the proposed tunable power parameter controlling the emphasis on relative error. In prior work by Sala et al. (2018), $p = 1$ was used implicitly, leading to uniform penalization of relative distortion.

However, in large hierarchical structures such as ours, the distribution of target distances is highly imbalanced: most node pairs are far apart (e.g., $\frac{203 \cdot 202}{2} = 20503$ pairs are at the maximum distance of 6 edges), while far fewer node pairs are closely related (e.g., *subparts* of the same *part*, with a distance of 2 between them). This imbalance biases the optimization to prioritize global accuracy at the expense of local separation, which is suboptimal for classification tasks where errors frequently occur between semantically close classes.

To address this, we improve the method by introducing the power parameter p and setting it to values greater than 1 to amplify high-magnitude errors. This indirectly changes the balance between local and global distortion: at low p , few large distortions, such as between *subparts* of one *part*, are less penalized than numerous small distortions, such as between *subparts* of different *wholes*. At higher p , the few large distortions become progressively more impactful on the loss, encouraging the optimization algorithm to distribute the relative error more uniformly across all pairs. We discuss the impact of this parameter in Section 7.1.2.

4.2 Tractable Prototypical Learning for Dense Tasks

Having established a method for constructing class prototypes, we now detail our classification mechanism. Instead of learning complex decision boundaries within the hyperbolic space, as is done in Hyperbolic MLR, our method uses prototypical learning. This approach allows us to leverage the inherent geometric properties of hyperbolic spaces without trainable parameters on the manifold itself, and thus avoid the complications experienced by the hyperbolic MLR.

In prototypical learning, we define vectors that act as archetypal representations for classes and enable a meaningful geometric organization of class semantics. By positioning similar or related classes in proximity within the embedding manifold, we can explicitly encode prior se-

semantic relationships, facilitating improved discriminability and hierarchical awareness in classification outcomes. A prediction is made by computing the distance of the hidden representation to all class prototypes and selecting the class corresponding to the closest vector.

However, applying this concept to a dense prediction task like semantic segmentation introduces a significant computational challenge, posed by the standard formulation of the hyperbolic distance in the Poincaré ball model:

$$d(x, y) = \frac{2}{\sqrt{-c}} \tanh^{-1} (\sqrt{-c} \| -x \oplus_c y \|) .$$

In tasks like segmentation, this expression must be evaluated for every pixel representation against a set of class prototypes in hyperbolic space. In our case, this entails computing distances between $128 \times 128 \times \text{batch_size}$ pixel vectors and over 200 class prototypes, resulting in prohibitive memory and compute demands using standard implementations.

4.2.1 Reformulation of Hyperbolic Distances

To address this, we propose an optimization of this computation inspired by Atigh et al. (2022). First, note that the hyperbolic distance function only requires the *norm* of the Möbius difference $-x \oplus_c y$, not the resulting vector itself. Using the definition of Möbius addition from Equation (3.2), let us denote the numerator of that expression by **num** and the denominator by **denom**. Note that while the numerator is a vector in $\mathbb{D}^n$, denominator is a scalar. Then:

$$\| -x \oplus_c y \| = \sqrt{\sum_i \left(\frac{\mathbf{num}_i}{\mathbf{denom}} \right)^2} = \frac{\sqrt{\sum_i \mathbf{num}_i^2}}{\mathbf{denom}} .$$

To compactly express repeated terms, we define:

$$\alpha = 1 - 2c\langle x, y \rangle - c\|y\|^2, \quad \beta = (1 + c\|x\|^2).$$

and obtain

$$\begin{aligned} \mathbf{num}_i^2 &= (\alpha(-x_i) + \beta y_i)^2 \\ &= (\alpha^2 x_i^2 - 2\alpha\beta x_i y_i + \beta^2 y_i^2) \end{aligned}$$

Expanding the squared sum, we can compute the scalar directly:

$$\sum_i \mathbf{num}_i^2 = \|\mathbf{num}\|^2 = \alpha^2 \|x\|^2 - 2\alpha\beta \langle x, y \rangle + \beta^2 \|y\|^2 .$$

This representation requires computing only 3 vector operations: $\|x\|^2$, $\|y\|^2$, $\langle x, y \rangle$, and thus avoids instantiating any high-dimensional intermediate tensors when x and y are large matrices, which reduces the computational cost by several orders of magnitude for batch computations

of the norm. The final hyperbolic distance is then obtained by:

$$d(x, y) = \frac{2}{\sqrt{-c}} \tanh^{-1} \left(\sqrt{-c} \frac{\sqrt{\alpha^2 \|x\|^2 - 2\alpha\beta \langle x, y \rangle + \beta^2 \|y\|^2}}{1 - 2c \langle x, y \rangle + c^2 |x|^2 |y|^2} \right).$$

This implementation is mathematically equivalent to the standard formulation, but requires drastically fewer resources. In practice, it enables the computation of all pairwise distances between a large batch of feature vectors and class prototypes. This unlocks the feasibility of applying hyperbolic prototypical learning to dense, high-resolution tasks like semantic segmentation.

4.2.2 Treating Distances as Logits

With this efficient distance computation in place, our full prototypical learning pipeline operates as follows. First, the feature vectors produced by the network’s decoder head for each pixel are projected into the Poincaré ball using the exponential map from [Equation \(3.1\)](#).

While classification is technically performed by finding the nearest class prototype for each pixel’s hyperbolic embedding, we aim to integrate this geometric comparison into a standard training loop with cross-entropy loss. To that end, we convert the computed distances into logits: for a pixel at location (i, j) and a class l , the logit is defined as the scaled negative hyperbolic distance:

$$\text{logits}_{ijl} = -\tau \cdot d_c(r_{ij}, p_l),$$

where d_c is the hyperbolic distance under curvature c , r_{ij} is the hyperbolic representation of the pixel, p_l is the fixed prototype for the class, and τ is a temperature hyperparameter that scales the logits, controlling the sharpness of the final probability distribution.

This integrated approach provides several key advantages after training. Most importantly, it decouples the feature encoder from the classification mechanism. The network’s parameters are optimized to produce geometrically meaningful embeddings, while the classification rule is a fixed, non-learnable comparison to a set of precomputed prototypes. This separation provides remarkable flexibility: the prototypes can be modified, swapped, or extended post-training without the need for retraining any part of the model. As such, the method is fundamental to the unique capabilities of our model, namely multi-level generalization and remarkable zero-shot segmentation, which are explored in subsequent sections.

4.3 Shared Encoder Model

When predicting various levels of a single hierarchy, we are inevitably dealing with the same space of input images, as only the labels are different. Thus, it is computationally desirable, and practically suitable, to use a shared encoder for the images, with separate prediction heads for each level of the hierarchy.

To this end, we initialize the transformer encoder from a pretrained general segmentation

model and create 3 copies of the prediction head. Each head is then configured for its task: it obtains the set of prototypes that corresponds to its prediction level, and randomly initializes layers that require different dimensions. The schematic overview of this architecture is presented in [Figure 4.2](#). During training, the loss is computed separately for each granularity and then added together before the backward pass.

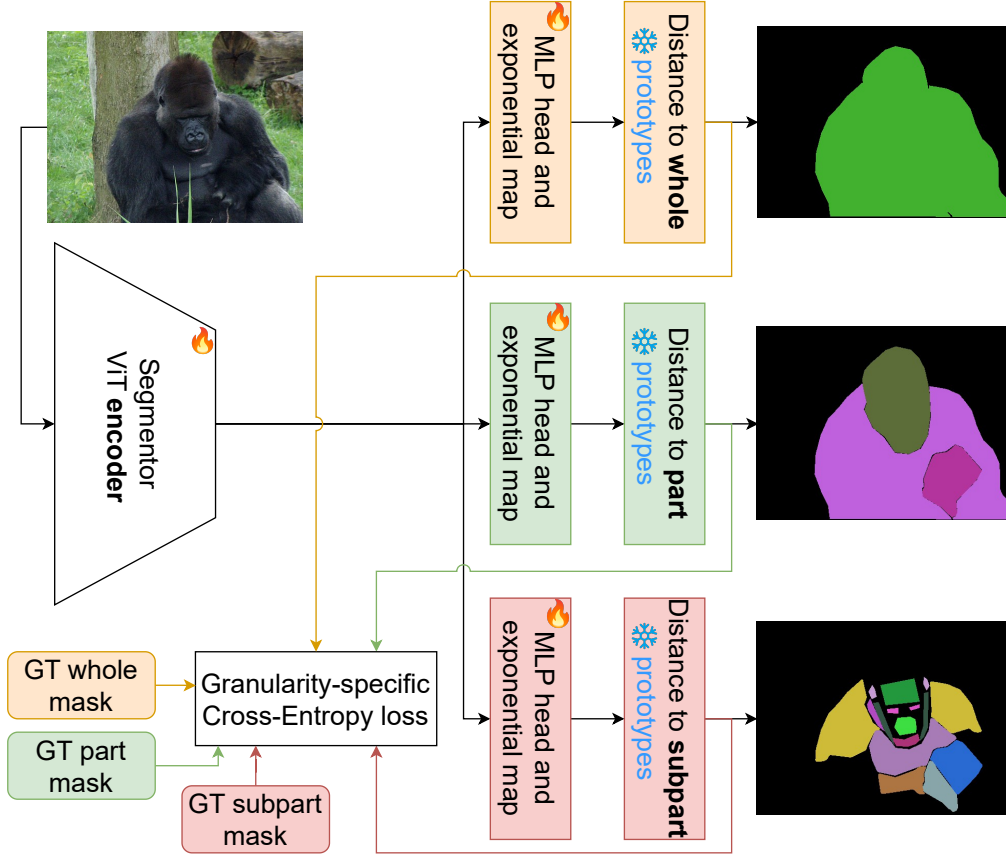


Figure 4.2: **Hyperbolic shared encoder architecture.** An image is processed by a transformer encoder to extract features, which are then passed to three decoding heads corresponding to different levels of the part-whole hierarchy. Each head uses distinct prototypes and is supervised independently.

4.4 Enabling Multi-Level Prediction

A key advantage of our prototypical framework is the explicit encoding of the class hierarchy within the geometry of the embedding space. This enables capabilities beyond standard segmentation, such as producing segmentation masks for higher hierarchical levels, entirely without the corresponding labels.

To achieve this, we first construct the prototypes for all classes in the hierarchy as described in [Section 4.1](#). Then, we train a full model (encoder + one head) with masks exclusively for the most detailed available granularity: the leaves of the hierarchy. Crucially, during inference, we can extend this same mechanism to predict masks at higher levels of the hierarchy. Because the prototypes for coarser levels are embedded in the same hyperbolic space, we can compute

distances from the same pixel representations to prototypes at all granularity levels. This results in a segmentation output at multiple levels using a single forward pass of the network. These predictions require no additional supervision beyond the *subpart*-level labels used during training.

This design allows our model to produce coarse-to-fine segmentations without training on any explicit annotations for the coarser levels. As shown in Figure 4.3, all three segmentation masks (*whole*, *part*, and *subpart*) are generated from the same encoder outputs, highlighting the flexibility and generality of the proposed framework.

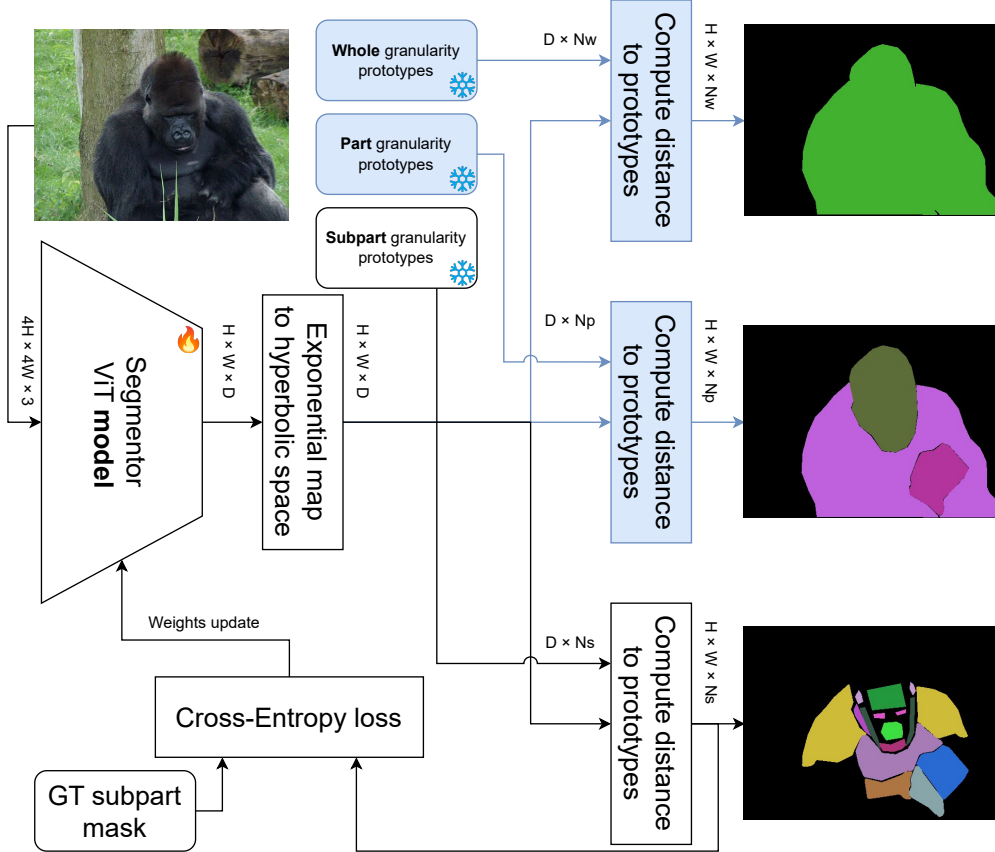


Figure 4.3: **Hyperbolic prototypical multi-level prediction.** Our framework generates predictions for all three hierarchical levels, despite being supervised on only one. The components active exclusively during this inference-time generalization are highlighted in blue. D is prototype dimensionality. N_w , N_p and N_s are the amount of classes for granularities *whole*, *part*, and *subpart* respectively.

4.4.1 Zero-Shot Evaluation

Our approach leverages the flexibility of structuring class prototypes in embedding spaces, which can introduce additional inductive bias compared to traditional classification setups. To evaluate the zero-shot capabilities of our segmentation model, we adopt a specific experimental procedure on our primary dataset, SPIN.

In the zero-shot scenario, we select one of the top-level categories in the given class hierarchy, such as *Quadruped*, to serve as the unseen category. The training protocol remains similar to

the standard training regime but with crucial filtering: during training, we exclude any images containing pixels from the selected zero-shot category. Consequently, the model is never exposed to these classes in training. During evaluation, conversely, we exclusively assess performance on images containing the zero-shot category, ensuring predictions are limited to either the background class or the subclasses of the zero-shot *whole*.

Formally, we define a set of indices corresponding to the zero-shot classes, Z . At training time, images X_{train} are filtered such that no image contains pixels from any $z \in Z$. At evaluation time, images X_{eval} are chosen if and only if they contain pixels from categories in Z . The network is constrained to predict either background or classes from Z , thus strictly evaluating its capability to generalize from known hierarchical structures and learned representations to previously unseen categories.

Chapter 5

Experimental Setup

Our experiments relied on the SPIN (Myers-Dean et al., 2024) dataset for training. As we elaborate on in Section 5.3, this dataset makes it possible to construct a model that predicts segmentation masks at three levels simultaneously, which this thesis aims to explore.

To perform the experiments, we first construct a directed acyclic graph representing the hierarchy of the SPIN dataset. Starting from the root node, we add edges to each of the 11 *wholes*, and the background class. Then, we attach the 40 *part* granularity nodes to their respective *wholes*. Finally, we add the 203 *subpart* granularity leaves to the corresponding *part* nodes. Therefore, we obtain a tree with a single representation for the background class, 256 nodes, and a maximum depth of 3. Applying the process specified in Section 4.1 to the constructed graph, we obtain the class prototypes for all granularity levels, organized on the Poincaré ball.

We selected the SegFormer model (Xie et al., 2021) as our backbone for its performance in similar segmentation tasks. For all experiments, we initialize the model with the pretrained weights provided by the authors, and leave structure and logic of the backbone unchanged.

We implement a custom head that is based on the original SegFormer head, but incorporates our methodology for hyperbolic and prototype-based learning. Crucially, we need to reinitialize the last convolutional layer of the head, as our number of classes is different from that of the original model. Moreover, to ensure fair comparison in some setups, we sometimes introduce a bottleneck layer that converts hidden pixel representations from the original 256 to a lower dimension directly before the classification layer. We do not add a nonlinearity after the new layer, as it serves strictly as a bottleneck.

We train all resulting model parameters with the AdamW optimizer and a polynomial learning rate schedule, in line with the original SegFormer training setup (Xie et al., 2021). The full code is available on GitHub¹.

¹<https://github.com/mikhail-vlasenko/hyp-segmentation>

5.1 Hyperparameters

The hyperparameters for our model were initially informed by previous research. Once the core experimental setup was frozen, the hyperparameters were adjusted based on extensive empirical evaluations to achieve better performance with the standard Euclidean model. These values were locked and reused for all configurations. The hyperbolic-only parameters were then tuned separately. We stress that such hyperparameter search leaves the hyperbolic models at a disadvantage, reducing potential bias and simplifying the transition to hyperbolic training configurations for future research.

For the initial ViT checkpoint, our experiments use the pretrained model SegFormer-B4, fine-tuned on ADE20K. Training occurs over 50 epochs with an effective batch size of 16 and an initial learning rate of 1.2×10^{-4} , employing a polynomial learning rate decay schedule with a power of 1.0. The hyperbolic embedding space has a curvature of -1 , and prototype-based classification approaches utilize a hyperbolic classification temperature (τ) set to 15.

We also apply standard data augmentation strategies to enhance model robustness: random cropping to 80% of the original image size in both spatial dimensions and random horizontal flipping with probability $p = 0.5$.

5.2 Implementation and Tools

Our experimental framework was implemented using PyTorch (Paszke et al., 2019) and Hugging Face Transformers (Wolf et al., 2020) for deep learning, while Geoopt (Kochurov et al., 2020) and HypLL (van Spengler, Wirth, & Mettes, 2023) libraries were used to handle most of the hyperbolic geometry operations.

Model training and evaluation were enhanced with the Lightning framework (Falcon & team, 2019), providing structured experimentation pipelines and easy extensibility. All experiments used global seeds 0, 1, 2, and 42. Experiment tracking and logging were carried out through Weights & Biases, contributing to reproducibility and enabling comprehensive analysis of model performance and hyperparameter impact.

5.3 SPIN Dataset

Hierarchical semantic segmentation, particularly at granular levels, has historically faced substantial challenges due to the scarcity of suitable datasets. The recently introduced SubPartImageNet (SPIN) dataset (Myers-Dean et al., 2024) represents a significant advancement in this area, being the first comprehensive hierarchical segmentation dataset providing extensive annotations for natural images at three granularity levels.

While previous datasets such as ADE20K (Zhou et al., 2017) and PartImageNet (J. He et al., 2022) facilitated segmentation tasks at object and part levels, SPIN uniquely extends the annotation depth to include detailed subpart granularity across diverse natural categories. Unlike synthetic datasets (Mo et al., 2019; Xiang et al., 2020), which often exhibit limited

generalization to natural image contexts (Y. Wang et al., 2024; Chłopowiec et al., 2024), SPIN annotates real-world images from ImageNet (Deng et al., 2009). The dataset offers over 102,000 annotated masks across 11 *whole*, 40 *part*, and 203 *subpart* categories. Such diversity and scale make it a robust foundation for research in hierarchical segmentation. There are a total of 8828 training images, 519 validation images, and 1040 test images.

5.4 SegFormer

Previous work in hierarchical or hyperbolic segmentation (Weber et al., 2024; Atigh et al., 2022; Li et al., 2022) relied on sophisticated CNN-based architectures such as DeepLabV3+ (L.-C. Chen et al., 2018) with ResNet-101 backbone (K. He et al., 2016) and OCRNet (Yuan et al., 2021) with HRNet-W48 backbone (J. Wang et al., 2020). While powerful, these models have been surpassed by modern alternatives. Particularly, after the success of Vision Transformer (ViT) (Dosovitskiy et al., 2020), hierarchical Transformer architectures such as Pyramid Vision Transformer (PVT) (W. Wang et al., 2021) and Swin Transformer (Liu et al., 2021) were developed for dense prediction tasks like segmentation.

This thesis leverages SegFormer (Xie et al., 2021), which uniquely combines a hierarchical Transformer encoder, termed Mix Transformer (MiT), with a lightweight all-MLP decoder. A key innovation in the MiT encoder is its ability to output multi-scale features without relying on explicit positional encodings. Instead, it employs overlapping patch merging to maintain local continuity, efficient self-attention mechanisms to reduce computational cost, and a Mix-FFN that incorporates 3x3 depth-wise convolutions to implicitly provide positional information. This design avoids the performance degradation often seen when interpolating fixed positional codes for varying test image resolutions. The model architecture is presented in Figure 5.1.

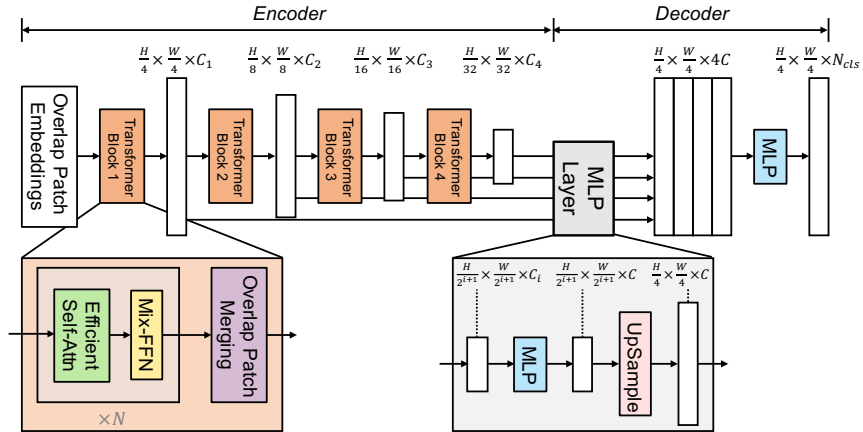


Figure 5.1: The SegFormer framework consists of two main modules: a hierarchical transformer encoder to extract coarse and fine features; and a lightweight all-MLP decoder to directly fuse these multi-level features and predict the semantic segmentation mask. “FFN” indicates a feed-forward network. Figure from original paper (Xie et al., 2021).

The lightweight all-MLP decoder complements the MiT encoder by effectively fusing features from different hierarchical stages. As demonstrated by the authors, the powerful representations

from MiT, which capture both local and global attention and exhibit larger Effective Receptive Fields (ERFs) than traditional CNNs, enable this simple decoder to achieve high performance with minimal parameters.

For our work, we selected the SegFormer-B4 variant. It achieves 0.503 mIoU (single-scale) on the ADE20K dataset with 64.1 million parameters. This represents a significant improvement in both performance and efficiency compared to prior architectures like DeepLabV3+/ResNet-101 (0.441 mIoU, 62.7M parameters) or OCRNet/HRNet-W48 (0.456 mIoU, 70.5M parameters). The SegFormer-B4’s vastly superior performance-to-parameter ratio, coupled with its robust feature representation capabilities, makes it a favorable foundational architecture for exploring part-whole segmentation in the hyperbolic space.

5.5 Hierarchy-Aware IoU

To evaluate segmentation performance in a hierarchy-aware manner, we designed a novel metric termed the Graph-Distance Soft Intersection-over-Union (Soft IoU). Unlike standard Intersection-over-Union (IoU), which treats all class predictions as equally distinct, our Soft IoU metric incorporates the hierarchical proximity between classes as defined by the graph distance between them.

Specifically, for each pixel, the Soft IoU assigns similarity scores based on the graph-based shortest-path distances between the predicted class and the ground-truth class. This approach is intended to reward predictions that, while incorrect in a strict sense, remain close within the hierarchical structure of the class taxonomy. Thus, predictions with minor hierarchical deviations receive partial credit, capturing nuanced distinctions overlooked by standard IoU.

Mathematically, the standard IoU can be described as:

$$\text{IoU}_c = \frac{\sum_{x \in X_c} \mathbf{1}\{\hat{y}_x = c\}}{|X_c| + |\hat{X}_c| - \sum_{x \in X_c} \mathbf{1}\{\hat{y}_x = c\}},$$

where X_c denotes the set of all pixel locations whose ground-truth label is class c , similarly, $\hat{X}_c$ denotes the set of all pixel locations predicted by the model as class c . For a given pixel x , $\hat{y}_x$ represents the predicted class at pixel x . Finally, $\mathbf{1}\{\hat{y}_x = c\}$ is the indicator function that evaluates to 1 when the predicted class $\hat{y}_x$ matches c , and 0 otherwise, so that $\sum_{x \in X_c} \mathbf{1}\{\hat{y}_x = c\}$ counts the number of correctly predicted pixels for class c .

The Soft IoU is then a generalization of the standard IoU with a continuous monotonic function $f : \text{GT} \times \text{prediction} \rightarrow \mathbb{R}_{\geq 0}$ in place of the indicator function:

$$\text{Soft IoU}_c = \frac{\sum_{x \in X_c} f(d(c, \hat{y}_x))}{|X_c| + |\hat{X}_c| - \sum_{x \in X_c} f(d(c, \hat{y}_x))}, \quad f(d) = \frac{1}{1 + d},$$

where $d(c, \hat{y}_x)$ denotes the graph distance between the ground-truth class c and the predicted class $\hat{y}_x$ in the class hierarchy, and $f(d) = \frac{1}{1+d}$ is a decay function mapping that distance to a similarity score. Consequently, $\sum_{x \in X_c} f(d(c, \hat{y}_x))$ accumulates partial credit for predictions that are hierarchically close to c , rather than requiring an exact match.

However, empirical evaluation across multiple model variants, both hierarchy-informed and hierarchy-agnostic, revealed that Soft IoU scores closely correlate with traditional IoU values. This strong correlation indicates that, despite theoretical differences in sensitivity to hierarchical errors, the practical errors produced by these models are similarly distributed. To support this, we present a confusion matrix in [Figure 7.3](#), where it is clear that the model confuses almost exclusively sibling classes.

This observation underscores the robustness of IoU as a general-purpose segmentation metric, so this work primarily relies on this metric to evaluate model performance.

5.6 Previous State-of-the-Art

The SPIN dataset paper (Myers-Dean et al., [2024](#)) evaluates a range of contemporary models to establish baselines for hierarchical segmentation performance. Specifically, the paper benchmarks foundational models including HIPIE (X. Wang et al., [2023](#)), PixelLLM (Xu et al., [2024](#)), LISA (Lai et al., [2024](#)), GLaMM (Rasheed et al., [2024](#)), Ferret (You et al., [2023](#)), CoGVLM (W. Wang et al., [2024](#)), Shikra (K. Chen et al., [2023](#)), and Kosmos2 (Peng et al., [2023](#)) across three hierarchical granularity levels: *objects*, *parts*, and *subparts*. The models’ performance generally declines with increasing segmentation granularity, revealing the difficulty of subpart segmentation.

Notably, the fine-tuned GLaMM variant (GLaMM-FT) achieves the highest performance among all fairly evaluated models. We note that the authors’ evaluation of the Segment Anything Model (Kirillov et al., [2023](#)) was aimed to obtain an upper bound, and thus operated under the unrealistic condition of ground-truth bounding box inputs. However, it is important to highlight that aside from being a 7B-sized model, GLaMM-FT requires a separate forward pass for each class to produce a segmentation mask for a single image. This significant computational limitation restricts its practicality, particularly for applications demanding efficiency and scalability.

At the time of writing, no published research outperforms GLaMM-FT reported by Myers-Dean et al. ([2024](#)) on core segmentation metrics, so we compare our results to this model.

Chapter 6

Results

We now present the results of our extensive evaluation, validating the effectiveness and advantages of our proposed hyperbolic prototypical segmentation framework. The evaluation is structured to highlight the comparative performance against established baselines, demonstrate the benefits of hierarchical embeddings, and illustrate key advantages of our approach.

6.1 Comparison to State-of-the-Art

Table 6.1 summarizes the performance of our hyperbolic model in comparison with the current state-of-the-art GLaMM-FT model evaluated on the SPIN dataset. Remarkably, our hyperbolic prototypical model achieves superior performance at finer granularity levels (*part* and *subpart*) with significantly fewer parameters than all previous methods. Specifically, at the subpart level, we achieve an mIoU of 0.2676, representing a 9% improvement over GLaMM-FT’s performance of 0.2456. We stress that this improvement is achieved despite the dramatically lowered computational cost: GLaMM-FT has approximately 100 times more parameters and requires significantly more forward passes to create a mask for one input image.

Model	Whole mIoU	Part mIoU	Subpart mIoU	# parameters
LISA [†]	0.8545	0.3136	0.1155	13B
GLaMM [†]	0.8631	0.4	0.11	7B
GLaMM-FT [†]	0.9108	0.6076	0.2456	3 × 7B
Hyperbolic shared (ours)	0.895 ± 0.001	0.677 ± 0.005	0.2676 ± 0.0035	64M + 3 × 3M

Table 6.1: Our compact hyperbolic shared encoder architecture achieves a new SOTA on the SPIN dataset, surpassing the multi-billion-parameter GLaMM-FT on the most challenging *part* and *subpart* levels. Results show mean ± standard deviation over 4 seeds. [†]Reported by Myers-Dean et al. (2024).

These results validate that a hyperbolic prototype-based method can produce superior segmentation performance across most of the hierarchical levels, while maintaining low parameter count and excellent computational efficiency. Qualitative examples of our model’s multi-level output are shown in Figure 6.1.

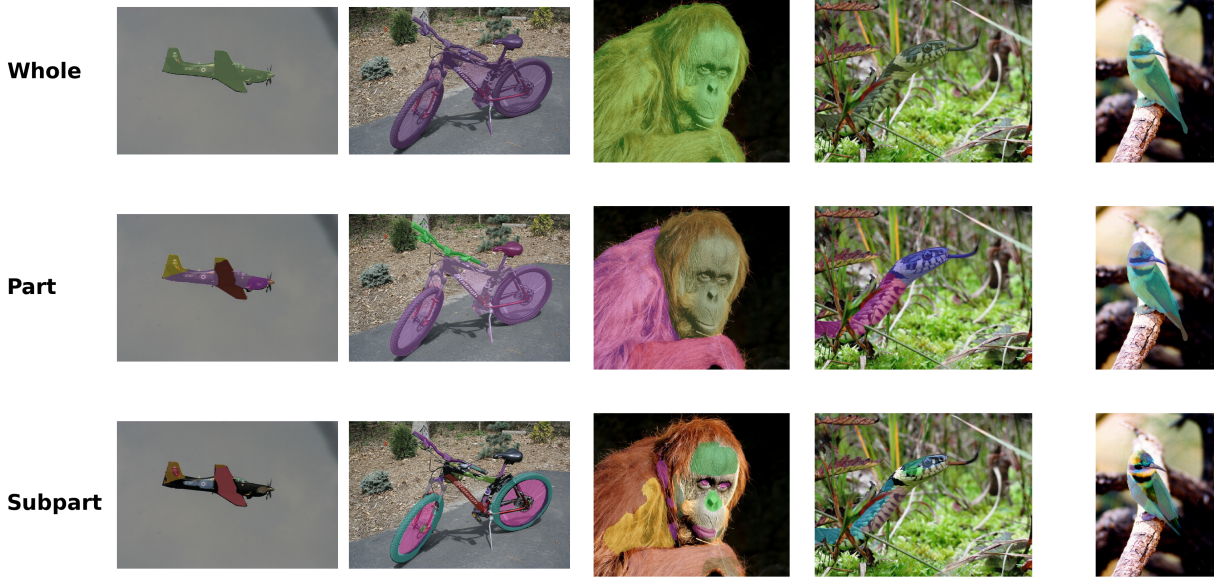


Figure 6.1: **Qualitative results** from our hyperbolic shared encoder model on the SPIN test set. Predicted masks are overlaid on the inputs. The results demonstrate the model’s ability to produce accurate and hierarchically consistent segmentations across diverse object categories.

6.2 Multi-Level Generalization

To evaluate the methodology proposed in [Section 4.4](#), we conducted an experiment where models were trained only on the *subpart* annotations and tested on their ability to predict *part* and *whole* masks, without any direct supervision on these coarser levels. The results are shown in [Table 6.2](#). The only difference between the two presented models is in the manifold of the prototype embeddings.

Manifold	Whole mIoU	Part mIoU	Subpart mIoU	Subpart soft mIoU
Euclidean	0.0672 ± 0.0004	0.1208 ± 0.0039	0.2100 ± 0.0054	0.2577 ± 0.0003
Hyperbolic	0.3874 ± 0.0070	0.3607 ± 0.0013	0.2300 ± 0.0036	0.3124 ± 0.0020

Table 6.2: Hyperbolic geometry enables **robust multi-level generalization**. When trained only on *subpart* masks, the hyperbolic model successfully predicts *part* and *whole* levels, while the Euclidean model fails in the same scenario. This highlights the strong inductive bias provided by the 8-dimensional prototypes.

Our hyperbolic model significantly outperforms its Euclidean counterpart in transferring hierarchical knowledge upward. This demonstrates our method’s strong inductive bias towards hierarchy-aware representation learning, crucially enabling robust multi-level prediction from limited granularity annotations.

Finally, we find that thanks to the advantages of the hyperbolic space, it is possible to embed the prototypes in $d \ll \text{num_classes}$ (such as 8 dimensions for 256 classes in this case), at negligible cost to segmentation performance. The downstream effect of the embedding dimension of the class prototypes is explored further in [Tables 6.3](#) and [7.1](#).

6.2.1 Annotation Challenges

While the hyperbolic model shows strong generalization, its performance on the coarser levels is naturally lower than that of a directly supervised model. This discrepancy is partially explained by the SPIN dataset’s annotation scheme. As illustrated in Figure 6.2, the ground-truth masks for subparts do not tile the area of their parent part entirely. Some of the pixels that belong to a *part* (e.g. the side wall of the camper van) may not be assigned to any specific *subpart*, meaning the model is taught to assign the background class to these pixels. Therefore, when making a *part*-level prediction on such pixels, the network is likely to misclassify them as the background.

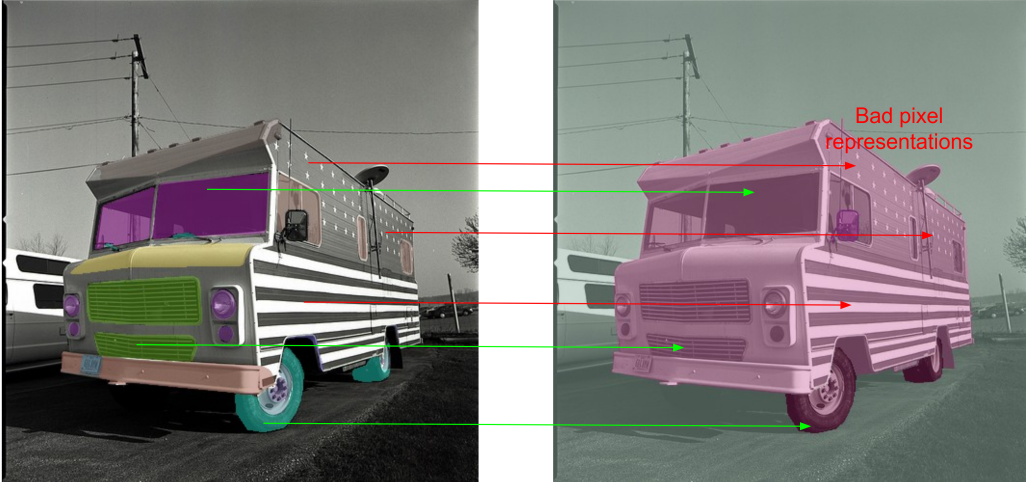


Figure 6.2: Annotation gaps in the SPIN dataset limit multi-level performance. **Left:** The ground-truth masks for the *subpart* granularity of a camper van. **Right:** The ground-truth mask for the corresponding *part* granularity.

6.3 Part-First Hierarchy

So far, we have only discussed a single class hierarchy, one where parts are direct ancestors of wholes that they belong to. However, we additionally propose a second class hierarchy for SPIN: one where similar parts across different wholes are generalized to whole-agnostic pseudo classes, and the whole-specific part classes are their direct children. A subtree of both hierarchies is presented in Figure 6.3 for visual comparison. Such hierarchy encourages the model to learn a more abstract, transferable concept of a *part* (e.g. ‘Torso’) that is not strictly tied to being part of a specific *whole* (e.g. ‘Bird’ or ‘Reptile’). Since the *part-to-subpart* relation in this hierarchy remains unchanged, we benchmark this new hierarchy on the *part* level, rather than the *subpart* level.

As presented in Table 6.3, we find that the Part-first hierarchy is superior to our Standard hierarchy for classification at part granularity in low dimensions. In high dimensions, the difference is statistically insignificant, as it is within 0.5σ . This is expected, as such setup positions 41 *part*-level class embeddings in a 64-dimensional space, removing any structural bias.

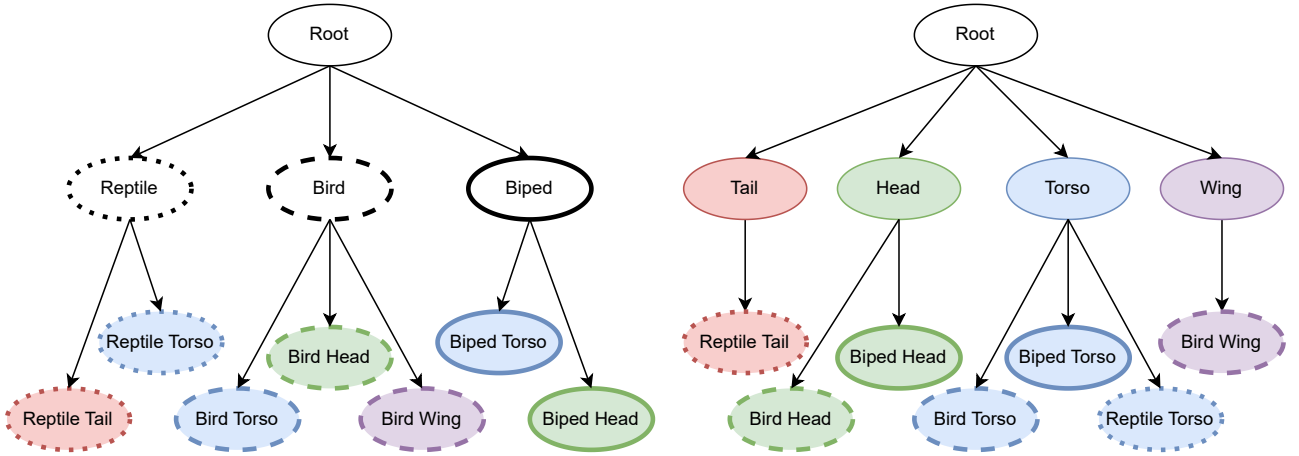


Figure 6.3: Subtrees of class hierarchies. **Left:** Standard hierarchy. **Right:** Part-first hierarchy. The Part-first hierarchy groups classes of the *part* granularity by their kind instead of the *whole* that they belong to. For example, heads of various animals become sibling classes.

Manifold	Hierarchy	Prototype dim.	Part mIoU
Euclidean	-	4	0.2517 ± 0.0581
Hyperbolic	Standard	4	0.6261 ± 0.0055
Hyperbolic	Part-first	4	0.6439 ± 0.0042
Euclidean	-	64	0.6778 ± 0.0006
Hyperbolic	Standard	64	0.6793 ± 0.0050
Hyperbolic	Part-first	64	0.6772 ± 0.0134

Table 6.3: Model segmentation performance for different hierarchies and dimensions of prototypes. Low prototype dimensionality of 4 shows the advantage of Part-first hierarchy; high prototype dimensionality of 64 produces equivalent results across hierarchies.

6.4 Zero-Shot Evaluation

Segmentation datasets commonly include a large portion of their classes in a single sample (Zhou et al., 2017; Lin et al., 2014; Cordts et al., 2016). This presents a challenge when attempting to perform a zero-shot evaluation, as filtering out all images containing the held-out class leads to a dramatic reduction in the training sample size. On the contrary, SPIN utilizes images from ImageNet (Deng et al., 2009), an object-centric classification dataset. Thus, each image contains strictly one type of the *whole* granularity class. This enables us to perform a genuine zero-shot evaluation rather than merely a zero-label scenario.

For this experiment, we select one of the 11 top-level categories (*wholes*) to be the unseen class (e.g. ‘Quadruped’). During training, we filter the dataset to exclude any image containing at least one pixel labeled as that *whole*. At test time, we evaluate the performance exclusively on images of that *whole*, and constrain the model to predict only the *parts* of that *whole*, or the background class. This setup measures the model’s ability to segment constituent parts of the held-out *whole* based on the hierarchical relationships learned from other entities.

As expected, the standard Euclidean model, lacking any explicit hierarchical structure, fails almost completely, achieving a mere 0.026 mIoU for the ‘Quadruped’ class. It has no mechanism to infer that the unseen ‘Quadruped Leg’ is semantically similar to a ‘Biped Leg’ that it has

Manifold	Hierarchy	Prototype dim.	Quadruped part mIoU	Bird part mIoU
Euclidean	-	4	0.0261 ± 0.0024	0.0268 ± 0.0010
Euclidean	-	16	0.0312 ± 0.0025	0.0296 ± 0.0010
Hyperbolic	Standard	4	0.1560 ± 0.0027	0.1458 ± 0.0014
Hyperbolic	Part-first	4	0.3834 ± 0.0052	0.3357 ± 0.0002
Hyperbolic	Part-first	8	0.3288 ± 0.0162	0.3588 ± 0.0084
Hyperbolic	Part-first	32	0.4122 ± 0.0273	0.3848 ± 0.0195

Table 6.4: Zero-shot performance on *part* granularity level with Quadruped or Bird as the held-out *whole*. Our hyperbolic prototypical method is vastly superior and achieves better mIoU at higher prototype dimensions.

seen during training.

Switching to a hyperbolic manifold with our standard prototype construction immediately yields a substantial improvement to 0.156 mIoU in an equivalent scenario. This demonstrates that the geometric prior — placing the unseen classes in the prototype space in a coherent manner, is a powerful signal for generalization.

To further improve this, we utilize the Part-first hierarchy. This modification proved exceptionally effective, boosting the zero-shot mIoU to an impressive 0.38 mIoU for the ‘Quadruped’ held-out *whole*. The model successfully learns to map the visual features of an unseen ‘Quadruped Leg’ to the general ‘Leg’ prototype cluster it formed during training. We observe that increasing the prototype embedding dimension further boosts the segmentation performance and achieves 0.41 mIoU with the highest tested dimension of 32. Notably, the trends and values of mIoU are highly consistent between the tested held-out *wholes*.

Finally, we note that the model’s zero-shot performance converges extremely quickly, achieving its best validation mIoU within the first 3 training epochs. This rapid convergence indicates that the model benefits only from the general alignment of its embedding space, rather than learning the fine-grained details of segmentation mask boundaries.

6.5 Evaluation on Pascal VOC

We further evaluate our approach on the Pascal VOC 2012 dataset (Everingham et al., 2010), which contains 20 object classes. From these, we map 6 classes directly to SPIN *wholes* (Aeroplane, Bicycle, Bird, Boat, Bottle, and Car) and group another 5 classes (Cat, Cow, Dog, Horse, Sheep) into the ‘Quadruped’ *whole*. Additionally, we align the Pascal VOC background class to the SPIN background category. This mapping strategy facilitates direct compatibility with our hyperbolic segmentation framework, ensuring consistency in evaluation criteria.

In this evaluation, the model is still trained exclusively on the SPIN training set, and the Pascal VOC dataset is utilized at evaluation time to assess out-of-distribution performance. To achieve this, we create a set of prototypes for the VOC classes by using a subset of the SPIN prototypes as described above. We then run the standard evaluation setup and compute distances to the VOC prototypes to make the prediction.

Results presented in [Table 6.5](#) show that the hyperbolic and Euclidean models achieve similar performance when evaluated on the Pascal VOC subset. The lack of significant differences indicates that, despite theoretical advantages of hyperbolic spaces for out-of-distribution scenarios, the benefits are not always noticeable in practice.

	SPIN whole mIoU	PASCAL VOC mIoU
Euclidean	0.8842 ± 0.0031	0.5576 ± 0.0082
Hyperbolic	0.8820 ± 0.0049	0.5528 ± 0.0038

Table 6.5: Generalization performance to PASCAL VOC 2012.

Chapter 7

Ablations

This chapter presents a series of ablation studies designed to dissect our framework and quantify the contribution of its key components. We evaluate our novel prototype construction technique, the influence of critical hyperparameters, and other significant design choices.

7.1 Prototype Construction

In this section, we discuss the effects of our proposed changes to the hyperbolic prototype construction protocol. We find that both of these changes provide substantial improvements to the segmenter’s performance.

7.1.1 Tree-Initialized Hyperbolic Prototypes

The proposed tree-aware initialization strategy was designed to prevent the optimization process from settling into suboptimal geometric configurations. We compare it against the standard random initialization baseline to quantify its effect on both the final prototype structure and the downstream segmentation performance.

The visualizations in [Figure 7.1](#) provide a stark qualitative contrast between the two approaches. The prototypes optimized from random initialization ([Figure 7.1a](#)) exhibit a highly irregular arrangement. The optimizer struggles to correct the initial poor topology, resulting in a final embedding with higher distortion and less meaningful geometric relationships between classes.

In contrast, when using our tree-aware initialization, the optimization process produces a clear and interpretable structure. The root nodes are positioned near the center of the Poincaré ball, with their children radiating outwards, visually representing the parent-child relationships. This structured starting point provides a strong inductive bias, guiding the optimizer towards a more globally optimal solution that accurately represents the hierarchy ([Figure 7.1b](#)).

The most critical outcome of this improved geometric arrangement is its direct impact on the downstream segmentation task. A well-structured prototype space, where distances between prototypes reflect the graph distance, simplifies the classification task for the model. As

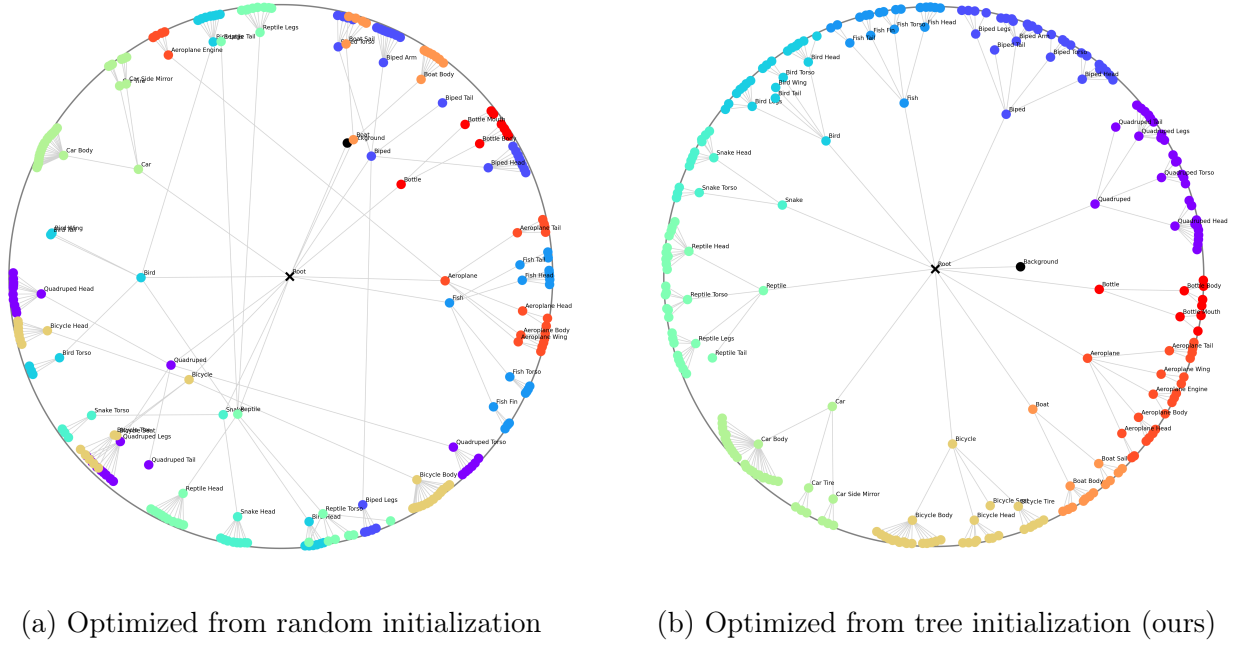


Figure 7.1: The impact of our tree-aware initialization on the final prototype geometry. Our method results in a much more hierarchically coherent structure.

shown in Table 7.1, models using tree-aware initialization consistently and significantly outperform those with random initialization across various embedding dimensionalities. The benefit is particularly pronounced in lower-dimensional spaces, which are more sensitive to poor topology. For instance, in an 8-dimensional space, tree-aware initialization achieves a subpart mIoU of 0.2300, a substantial 15% relative improvement over the 0.1992 achieved with random initialization. This confirms that providing the optimization with a strong, structurally-informed starting point is crucial for learning effective and geometrically coherent class representations, which in turn increases classification accuracy.

7.1.2 Loss Power

Initialization	Power p	Prototype dim.	Subpart mIoU
Random	1	64	0.1544 ± 0.0007
Random	3	4	0.1628 ± 0.0010
Random	3	8	0.1992 ± 0.0028
Random	3	64	0.2166 ± 0.0022
Random	5	64	0.2163 ± 0.0023
Tree-aware	1	64	0.1676 ± 0.0051
Tree-aware	3	4	0.0973 ± 0.0012
Tree-aware	3	8	0.2300 ± 0.0036
Tree-aware	3	64	0.1843 ± 0.0037

Table 7.1: Prototype construction methods significantly improve segmentation performance. The combination of our proposed tree-aware initialization and a distortion loss with power $p = 3$ yields the highest subpart mIoU, demonstrating the effectiveness of both contributions.

We now ablate the distortion- p loss modification, as described in [Section 4.1.2](#). This modification was motivated by the observation that in a large hierarchy, the vast majority of class pairs are distant, potentially biasing the standard distortion loss ($p = 1$) to optimize for global structure at the expense of local separation. For a classification task, however, accurately representing the small distances between semantically similar classes is paramount, as these are the most likely to be confused.

We trained our hyperbolic model using prototypes generated with different values of the power parameter p . As presented in [Table 7.1](#), the downstream mIoU is found to be highest for $p = 3$, so we use this value for all other experiments.

[Figure 7.2](#) plots the difference in relative distortion between embeddings trained with $p = 3$ and the standard $p = 1$. Nodes indexed from 1 to 203 are the prototypes of *subpart* granularity, and we can clearly observe that distortion is substantially reduced along the diagonal, where the absolute distance between nodes is smallest. This confirms our hypothesis: increasing the power p successfully rebalances the optimization to prioritize local accuracy.

The practical benefit of this rebalancing becomes evident when we consider the model’s typical error patterns, illustrated by the confusion matrix in [Figure 7.3](#). The most frequent errors occur between siblings in the hierarchy. Visually, these errors are represented by the squares of various sizes along the diagonal. By enforcing a more accurate geometric separation between these easily confused classes, the $p = 3$ loss enhances the model’s ability to discriminate between them, directly leading to a reduction in classification errors.

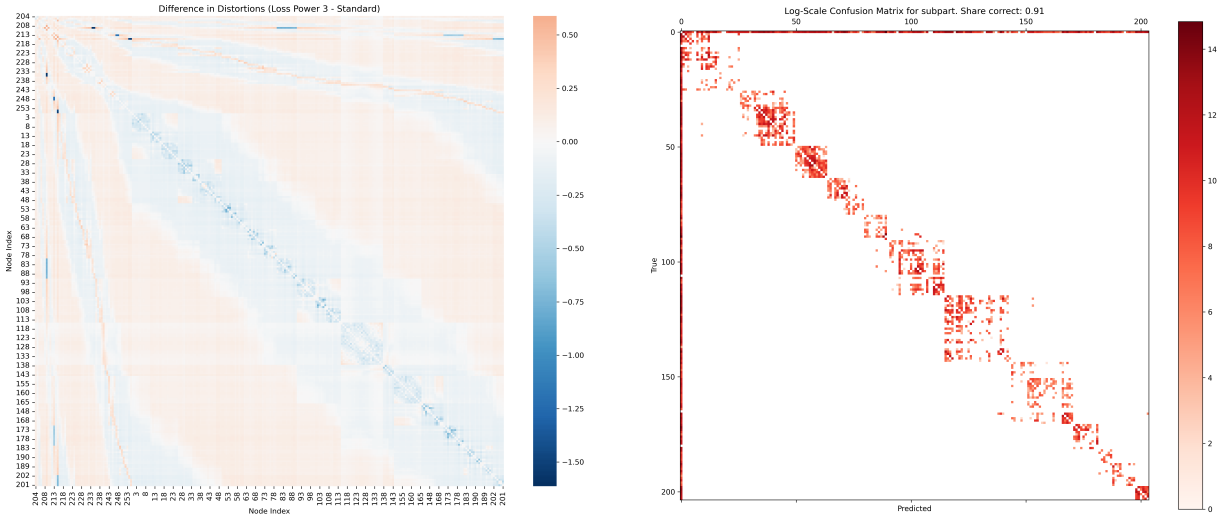


Figure 7.2: Difference in distortion for the power = 3 and the standard power = 1.

Figure 7.3: Typical confusion matrix for the *subpart* level.

7.2 Tuning Prediction Temperature

A critical hyperparameter in our prototypical learning framework is τ , the scaling factor for the negative hyperbolic distances that form the prediction logits. As described in [Section 4.2.2](#), this parameter effectively acts as the temperature for the final softmax distribution, controlling

the sharpness and confidence of the model’s predictions. A higher τ leads to a sharper, more peaked distribution, while a lower τ results in a softer, more uniform distribution over the classes.

The results, presented in Figure 7.4, reveal an important trade-off. For the *subpart* level, on which the model is directly supervised, performance peaks at $\tau = 15$. This indicates that a relatively high temperature is beneficial for distinguishing between the 203 fine-grained *subpart* classes.

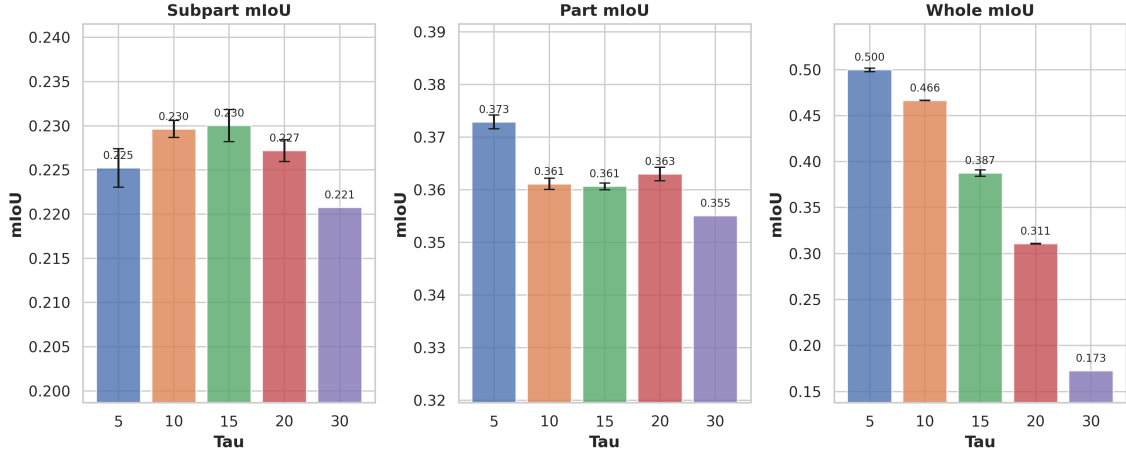


Figure 7.4: The effect of prediction temperature τ on multi-level generalization performance. While the directly supervised *subpart* task performs best at a medium temperature, the generalized *part* and *whole* predictions are optimal at a lower temperature. Same experimental setup as in Section 6.2

In stark contrast, for the hierarchically generalized *part* and *whole* levels, performance is optimal at the lowest tested temperature, $\tau = 5$, and degrades significantly as τ increases. Since a single model must be trained with a single value of τ , a compromise is unavoidable. For our main experiments, we selected $\tau = 15$, as it optimizes performance on the most granular supervised task.

Temperature for Euclidean manifold In a similar manner, we experimented with rescaling for Euclidean prototypes, but we observed no noticeable differences in the explored range.

7.3 Centered Background Class

One design choice in our hierarchy construction is to place the single background class on the same level as the *whole* granularity classes. Alternatively, we could use it as the root of the hierarchy, placing the prototype closest to the center of the Poincaré ball, and setting the *whole* classes as its children. Our tests show that positioning the background in the root node reduces the *subpart* mIoU by 5% (a statistically significant 5σ deviation). Therefore, we do not use it for other experiments. This finding is consistent with previous work by Atigh et al. (2022), who determined that the center of the hyperbolic ball corresponds to a prediction with low confidence.

Chapter 8

Conclusion

This thesis challenged the conventional paradigm in semantic segmentation, which typically treats classes as an independent and unstructured set, by developing a novel framework that embeds class hierarchies directly into the geometry of a hyperbolic space. We demonstrated that explicitly modeling hierarchical relationships provides significant advantages over traditional approaches.

Our hyperbolic prototypical model set a new state-of-the-art mIoU on *subpart* and *part* levels of the SPIN dataset, achieving this with two orders of magnitude fewer parameters and significantly more computationally efficient inference than the previous leading method. This success was underpinned by our novel contributions to prototype construction and architecture, including a tree-aware initialization and a distortion- p loss, which together create more effective class representations. The true power of this geometrically-informed approach was most evident in its ability to generalize. The model successfully predicted coarse-level segmentations from fine-grained training and achieved remarkable zero-shot performance on unseen object categories, tasks where equivalent Euclidean models failed. These capabilities were made practical for dense prediction through a critical optimization of the hyperbolic distance computation, marking the first successful application of this technique to a segmentation task.

8.1 Limitations

This work has several limitations. We observed that in some supervised settings, prototypes positioned in the maximal class separation configuration slightly outperformed those strictly adhering to the hierarchy, a counter-intuitive finding that warrants further investigation. Moreover, the annotation scheme of the SPIN dataset, where *subpart* masks do not fully cover their parent parts, inherently limits the measured performance in our cross-level generalization experiments.

8.2 Future Work

The findings and limitations of our research open up several exciting avenues for future investigation. To address the dataset’s annotation gaps, a model could be trained with joint supervision, leveraging *part*-level labels for pixels that lack a specific *subpart* annotation. The strong generalization results also suggest that this framework is an excellent candidate for few-shot or weakly-supervised learning, such as predicting fine-grained subparts from only coarse *part*-level labels. Exploring methods to learn class hierarchies directly from data or allowing prototypes to adapt dynamically during training could further enhance the model’s flexibility and performance, moving us closer to a new generation of segmentation models that build a truly structured understanding of the visual world.

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